

RETAIL FRAUD DETECTION

Project Workflow Report

Dashboard Analysis| SQL(MS SQL Server) | Power BI

1. Project Overview

1.1 Project Objective

This project aimed to analyse 100,000 retail transactions to identify patterns, concentrations, and risk factors associated with fraudulent activity. The central question driving the analysis was:

"Which payment methods, device types, merchant categories, and geographic origins carry the highest fraud risk — and what do those patterns reveal about the nature of retail transaction fraud?"

1.2 Dataset Description

Attribute	Detail
Dataset size	100,000 transactions
Reporting period	May 2025 – March 2026
Total fraud transactions	47,508
Total fraud loss	\$6,960,000 (6.96M)
Average transaction amount	\$120.48
Dimensions available	Payment method, Device type, Merchant category, Country, Date
Dashboard tool	Power BI
Data format	Structured tabular data with filters (Country, Device, Date)

1.3 Tools & Technologies Used

- Power BI — dashboard creation, interactive visualisations, DAX measures
- Data source — structured transaction dataset with fraud labels
- Visualisation types used — bar chart, line chart, treemap, donut chart, KPI cards
- Filters applied — Country (All), Device (All), Date range (All)

02

Data Collection & Preparation

ETL steps and transformations

2. Data Collection & Preparation

2.1 Data Source

The dataset was sourced as a structured retail transaction file containing labelled fraud and non-fraud transactions. Each record represented a single transaction with associated metadata fields.

2.2 Data Cleaning Steps Using SQL

The following preparation steps were applied before analysis:

Step	Action Taken	Reason
1	Verified row count = 100,000	Confirm dataset completeness
2	Checked for null/missing values	Ensure no gaps in key fields
3	Validated date range (May 2025–Mar 2026)	Confirm temporal coverage
4	Confirmed fraud label column (binary)	Required for KPI calculations
5	Verified transaction amount field	Used for average calculation
6	Standardised category & device field values	Consistent grouping in visuals
7	Removed duplicate transaction IDs (if any)	Prevent double-counting

Q1. Which payment method has highest fraud?

101

84 %✔ No issues found

Results

Messages

	payment_method	Fraud_count
1	Credit Card	9518
2	Google Pay	9504
3	PayPal	9500
4	Apple Pay	9496
5	Debit Card	9490

Q2. Which device type is risky?

84 % ✓ No issues found

Results Messages

	device_type	fraud_cases
1	Mobile	17373
2	Tablet	15185
3	Desktop	14950

Q3. Which merchant category has most fraud?

84 % ✓ No issues found

Results Messages

	merchant_category	Fraud_cases
1	Fashion	8105
2	Luxury	7996
3	Gaming	7938
4	Groceries	7914
5	Travel	7781
6	Electronics	7774

Q4. International vs Domestic Fraud

203

204 -- Insight : International transactions generally carry more fraud risk.

84 % ✓ No issues found

Results Messages

	is_international	Fraud_cases
1	0	13289
2	1	34219

Q5. Fraud by Country

218

84 % ✓ No issues found

Results Messages

	location	Total_fraud_cases
1	Germany	8039
2	Australia	8006
3	Canada	7915
4	India	7884
5	USA	7865
6	UK	7799

Q6. High Risk Customers

232

233

--Insight : Top High-Risk Customer: Customer C29180 is identified as the highest-risk customer, having the greatest total fraud amount of 1,923.19.

84 %

No issues found

Results

Messages

	customer_id	fraud_transaction	Total_fraud_amount
1	C29180	5	1923.19
2	C25317	5	1907.8
3	C12359	6	1852.9
4	C28630	9	1804.04
5	C28695	4	1787.65
6	C02118	3	1779.32
7	C19186	3	1736.29
8	C11549	6	1713.04
9	C27868	8	1702.94
10	C02611	4	1663.99

Q7. Fraud Trend Over Time

245

246

-- Insight : the top 10 dates with the highest number of fraud cases, all range between 151 and 158 incidents.

84 %

No issues found

Results

Messages

	transaction_date	fraud_cases
1	2026-03-01	158
2	2025-07-19	157
3	2025-04-30	155
4	2025-08-26	154
5	2026-01-30	152
6	2026-04-29	152
7	2026-01-12	151
8	2026-03-22	151
9	2025-05-24	151
10	2025-11-23	151

Q8. Does velocity impact fraud?

256

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-- Insight :
--Non-High-Velocity Fraud: Transactions flagged as 0 (not high-velocity) account for 71,602 fraud cases.
--High-Velocity Fraud: Transactions flagged as 1 (high-velocity) account for 28,398 fraud cases.

84 %

No issues found

Results

Messages

	velocity_flag	fraud_cases
1	0	71602
2	1	28398

Q9. Does previous fraud history matter?

269

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-- Insight : Higher Risk with History: accounts flagged with previous_fraud_flag = 1 (those who have committed fraud before) account for 34,337 fraud cases.

--Lower Risk without History: accounts with previous_fraud_flag = 0 (those with no known history) account for only 13,171 cases.

84 %

▼

✔ No issues found

Results

Messages

	previous_fraud_flag	Fraud_cases
1	0	13171
2	1	34337

Q10. Average Fraud Transaction Amount

84 %

▼

✔ No issues found

Results

Messages

	AVG_fraud_transaction
1	146.43

Q11. Rank highest fraud customers

Results

Messages

	customer_id	fraud_amount	fraud_rank
1	C29180	1923.19	1
2	C25317	1907.8	2
3	C12359	1852.9	3
4	C28630	1804.04	4
5	C28695	1787.65	5
6	C02118	1779.32	6
7	C19186	1736.29	7
8	C11549	1713.04	8
9	C27868	1702.94	9
10	C02611	1663.99	10

Q13. Fraud Rate by Merchant Category

Results

Messages

	merchant_category	fraud_rate
1	Fashion	47.980000000000
2	Luxury	47.720000000000
3	Gaming	47.480000000000
4	Travel	47.470000000000
5	Groceries	47.390000000000
6	Electronics	46.990000000000

Q14. Risk Score Analysis

Results		Messages
	fraud_risk	Risk_transactions
1	Medium	52211
2	Low	32203
3	High	15586

3.2 Visualisation Rationale

Each chart type was selected to best represent the underlying data structure:

Visualisation	Chart Type	Why This Type
Fraud by Payment Method	Horizontal bar chart	Ranked comparison across 5 discrete categories
Fraud Trend Over Time	Line chart	Continuous temporal data - shows change and volatility
Fraud by Device Type	Donut chart	Part-to-whole proportion across 3 categories
Fraud by Merchant Category	Treemap	Relative size comparison of 6+ categories simultaneously
International vs Domestic	Pie chart	Simple binary split - two-category proportion
KPI cards	Metric cards	Instant at-a-glance headline numbers for stakeholders

2.3 Derived Metrics & Calculated Fields

The following measures were created within Power BI using DAX:

- Fraud Transactions - COUNT of records where fraud label = TRUE
- Total Transactions - COUNT of all transaction records = 100,000
- Fraud Rate - Fraud Transactions / Total Transactions = 1.00 (displayed as scaled index)
- Fraud Loss Amount - SUM of transaction amount where fraud = TRUE = \$6.96M
- Average Transaction Amount - AVERAGE of all transaction amounts = \$120.48
- Fraud % by Dimension - Fraud count per group / Total fraud count × 100

